

ANDREW BRAVEN

FULL-STACK SOFTWARE DEVELOPER | MECHATRONICS ENGINEERING STUDENT

Nairobi, Kenya • +254 115 273 097 • amulibraven254@gmail.com • github.com/bonBavo

PROFILE

Full-Stack Software Developer & Mechatronics Engineering student with expertise in Java/Spring Boot backends, modern React/Next.js frontends, and networking fundamentals. Specialized in building scalable backend systems with microservices, event-driven architecture, and real-time communication. Strong cybersecurity foundations and currently learning game development with Unity. Proficient with TanStack ecosystem for advanced state management and data handling. Experienced with Kafka, Redis, Elasticsearch, and relational databases.

TECHNICAL SKILLS

Backend & APIs	Java, Spring Boot, Spring Security, REST APIs, GraphQL, Node.js, Express.js, Microservices
Frontend & Modern Web	React, Next.js, TypeScript, TanStack Query, TanStack Table, TanStack Router, Tailwind CSS, shadcn/ui
Networking & Real-Time	WebSockets, STOMP, Socket Programming, TCP/IP, HTTP/HTTPS, MQTT, Real-time systems
Databases & Search	PostgreSQL, MySQL, MongoDB, Elasticsearch, Prisma ORM, Data modeling, Schema design
Cybersecurity	Auth & Authorization, JWT, Secure APIs, Cryptography, OWASP, Network security, Secure coding
Distributed Systems	Apache Kafka, Redis, Event-driven architecture, Microservices patterns
DevOps & Tools	Linux, Docker, Git, GitHub, Maven, Gradle, Postman, OpenAPI/Swagger, JetBrains IDEs
Game Dev & Learning	Unity (currently learning), C#, C/C++, Blender, 3D modeling, ESP32, STM32, MQTT, IoT

FEATURED PROJECTS

BonRaccoon Studios — Full-Stack Platform

Full-Stack Developer • TanStack Start • Spring Boot • React • TypeScript • PostgreSQL • JWT • WebSockets

- Frontend: Built with React & TanStack ecosystem for modern state management and data handling.
- Backend: Designed Spring Boot REST API with JWT security, role-based access control, and microservices patterns.
- Networking: Integrated real-time WebSocket communication for studio updates and game showcases.
- Security: Comprehensive admin dashboard with secure authentication and content management.
- Applied networking principles for robust client-server communication and real-time data flow.

Full-Stack E-Commerce Application (Next.js)

Full-Stack Developer • Next.js • React • TypeScript • Prisma • PostgreSQL • Tailwind CSS

- Built adaptable e-commerce platform with dynamic content and configurable settings.
- Implemented full-stack architecture using Next.js server and client components.
- Applied TanStack patterns for efficient data fetching and client-side state management.
- Integrated Prisma ORM for type-safe database queries and schema management.
- Designed for easy configuration and reusability without requiring rebuilds.

E-Commerce Backend System

Backend Developer • Spring Boot • MySQL • Redis • Spring Security • Networking

- Developed complete backend with secure authentication and role-based access control.
- Implemented product catalog, order management, and protected admin endpoints.
- Designed REST APIs with validation, error handling, and network protocol optimization.
- Applied cybersecurity best practices for secure data transmission and access control.
- Optimized for scalability and performance with Redis caching layers.

Real-Time Chat Application Server

Backend Developer • Java • Spring Boot • WebSockets • Socket Programming • Networking

- Built real-time chat server using WebSocket and socket programming for low-latency communication.
- Managed concurrent client connections and reliable message routing over network protocols.
- Implemented secure message encryption and multi-factor user authentication.
- Applied networking principles for robust client-server architecture.
- Optimized for handling high-volume concurrent connections with excellent performance.

Nganya: Kenyan Transport Simulator

Lead Developer • Game Design • Systems Thinking • Simulation • Cultural Design

- Developing cultural prototype exploring authentic matatu/nganya transportation systems.
- Focused on realistic Kenyan urban transport mechanics and energy systems.
- Prototype-first approach with iterative refinement based on feedback.
- Planning to incorporate game development with Unity for 3D immersive experience.

EDUCATION

Murang'a University of Technology — BSc Mechatronic Engineering, In Progress

Relevant areas: Engineering Mathematics, Programming & OOP, Electronics, Instrumentation, Mechanics, CAD & Manufacturing, Mechatronic Systems

CERTIFICATION

MODCOM Institute — Full Stack Software Development Program

ADDITIONAL INTERESTS

Networking fundamentals & protocols • Cybersecurity engineering & network security • Backend architecture & system design • Distributed systems & microservices • Game development (learning Unity) • Full-stack development with React/Next.js/TanStack • Performance optimization & scalability • Real-time communication systems • Connected vehicles & IoT